Kingdom of Saudi Arabia

Ministry of Education

Computer Engineering

(471)

Project

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COE 326

Group 6776

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Introduction:

This lab introduces the basic instruments and components used in digital and microprocessor systems through a set of structured group experiments. Together with the group, these experiments were performed using the function generator to provide a variety of controlled test signals, and the oscilloscope to observe, measure, and analyze these signals in the time domain. The work also included studying latches and buffers as fundamental digital building blocks, examining how they store information and shape or isolate signals within larger digital circuits. Furthermore, the group explored how the microprocessor and memory cooperate to store, transfer, and process data as part of a complete digital system, linking the theoretical architecture to practical behavior on the bench. In addition, the group independently identified and used the relevant component data sheets while conducting the experiments, reflecting a careful and self-directed approach to preparing for and executing the laboratory work.

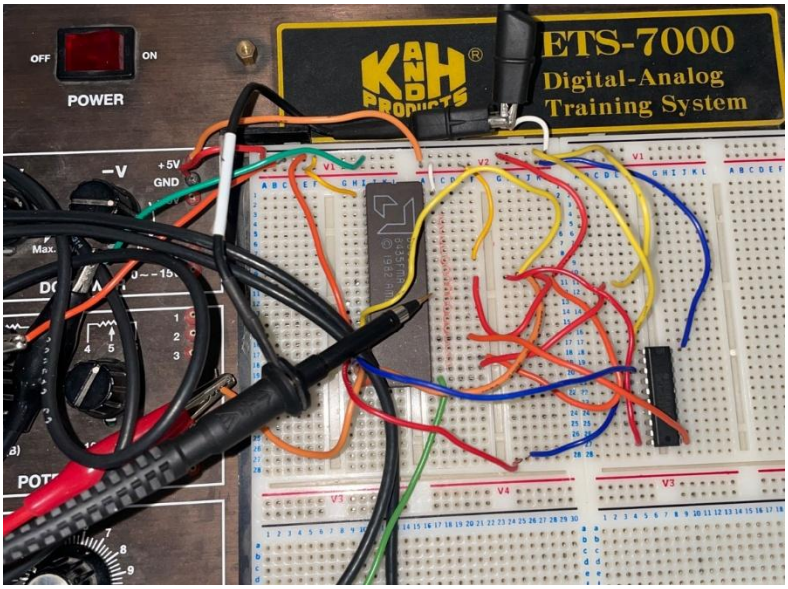
Experimental tools:

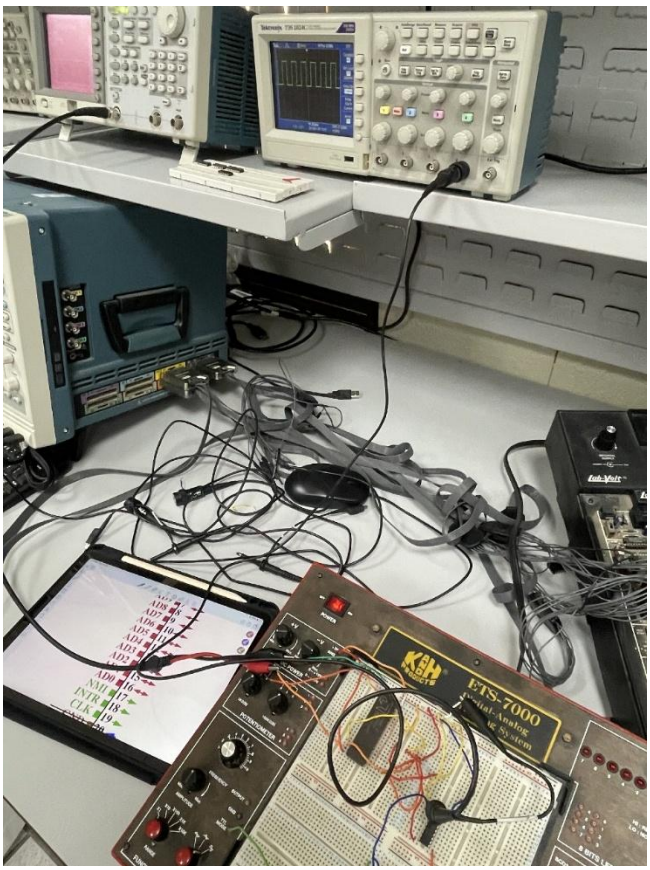
1. function generator
2. oscilloscope
3. latches
4. buffers
5. microprocessor
6. memory (The lack of time meant that we could not attach it)

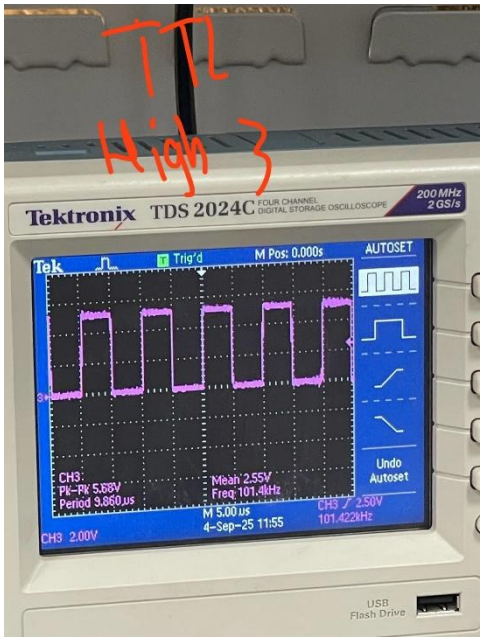
Datasheets

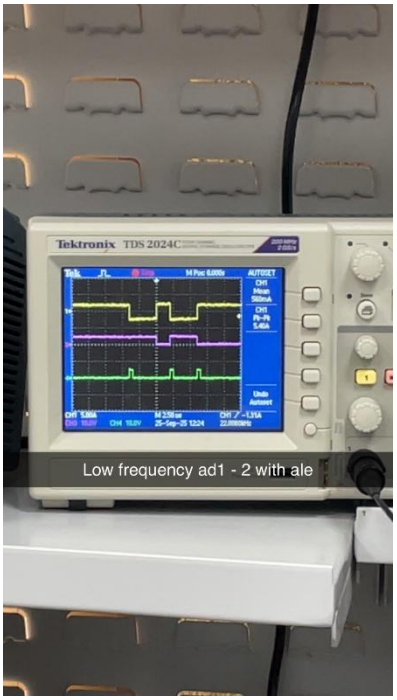
- Intel 8086 datasheet (16-bit HMOS microprocessor): https://datasheets.chipdb.org/Intel/x86/808x/datashts/8086/231455-006.pdf
- SN54HCT373 / SN74HCT373 octal transparent D-type latches (Texas Instruments): https://www.ti.com/lit/ds/symlink/sn54hct373.pdf
- SN74LS244 octal buffer/line driver (Texas Instruments): https://www.ti.com/lit/ds/symlink/sn74s244.pdf

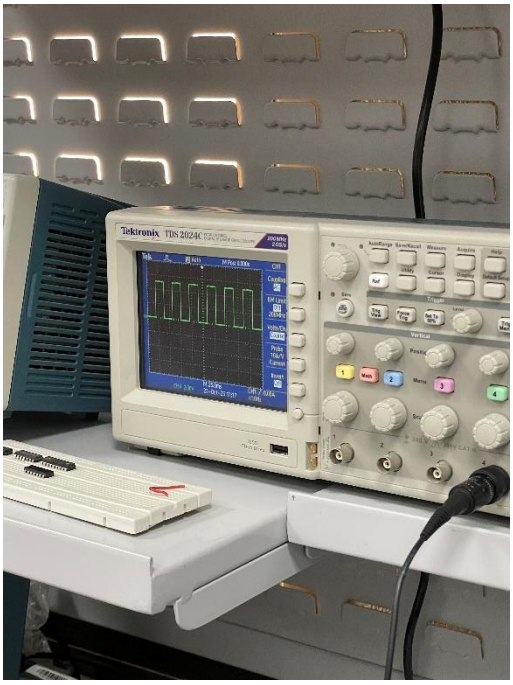
Latches testing:

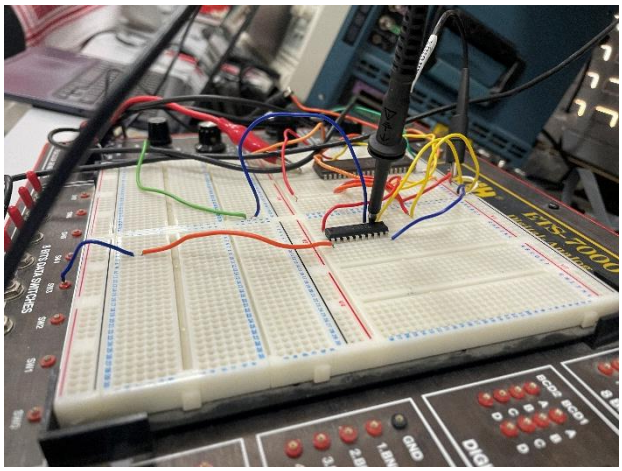


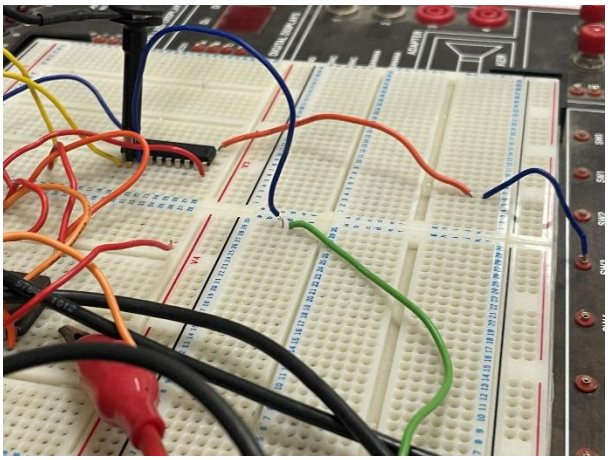


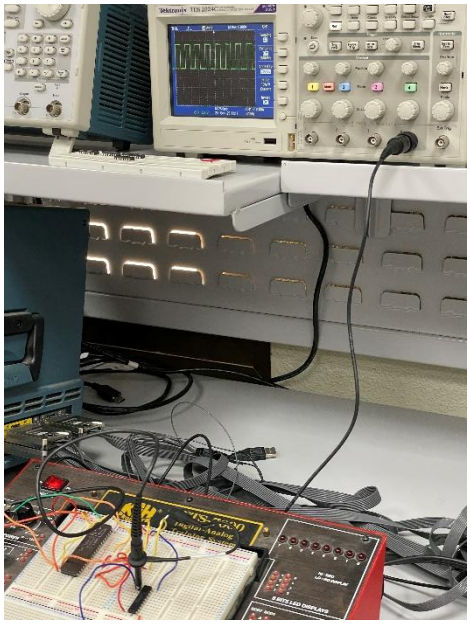

TTL
High 3
Tektronix TDS 2024C
FOUR CHANNEL DIGITAL STORAGE OSCILLOSCOPE
200 MHz 2 GS/s
Tek
Trig'd
M Pos: 0.000s
AUTOSET
CH3
Pk-Pk 5.68V
Period 9.860µs
Mean 2.55V
Freq: 101.4kHz
M 5.00µs
4-Sep-25 11:55
CH3 2.00V
CH3 ∕ 2.50V
101.422kHz
Undo Autoset
USB Flash Drive


Tektronix TDS 2024C
Low frequency ad1 - 2 with ale


Tektronix TDS 2024C







Latch Test Procedure:

- **Equipment Setup**

Connect the TTL latch IC (e.g., 74LS75 quad D-latch) to a breadboard with +5V VCC and ground. Use the function generator to supply enable ($\bar{G}$ or G) and data (D) inputs as square waves (0-5V, 1 kHz). Attach oscilloscope probes to D, $\bar{G}$, Q, and $\bar{Q}$ outputs with 10x attenuation for accurate TTL-level viewing.

Step 1: Power-On and Static Test

Apply power and verify all unused latch inputs are tied high or low per data sheet. Use switches or manual toggles to set D=0 or 1 with $\bar{G}$ low (latched), then toggle $\bar{G}$ high to observe transparent mode where Q follows D.

Step 2: Dynamic Enable Test

Set function generator for $\bar{G}$ as 50% duty cycle square wave. Apply steady D=1, trigger scope on $\bar{G}$ falling edge, and measure Q hold time after $\bar{G}$ goes low. Repeat with D=0 to confirm bistable operation and no glitches.

Step 3: Data Transition Test

With $\bar{G}$ low (latched), toggle D from low to high using generator burst mode. Raise $\bar{G}$ briefly to capture transition, then measure setup time (D stable before $\bar{G}$ low) and propagation delay ($\bar{G}$ to Q change) using scope cursors.

Step 4: Timing Verification

Vary frequencies up to 10 MHz and check for hold/setup violations from data sheet (e.g., 20 ns setup for 74LS75). Record max toggle rate where Q remains stable without metastability.

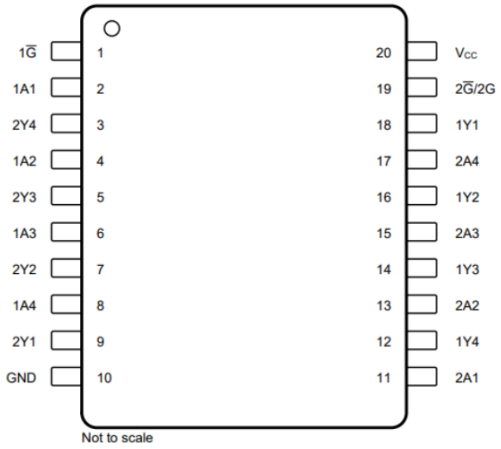
<u>Data Recording</u>

Tabulate measurements in a table: input waveforms, observed delays, and comparisons to data sheet specs. Note any loading effects or noise on breadboard.

Conclusion:

The latch test with TTL circuits verifies the basic storage and level-sensitive operation of D-type or SR latches, such as the 7475 or 74LS75 ICs, by applying logic inputs and observing output stability. Using a function generator to provide enable (G) and data (D) signals, and an oscilloscope to monitor Q and $\overline{Q}$ outputs, the experiment demonstrates how the latch captures and holds data when enabled while remaining transparent during the high-enable phase. Key measurements include propagation delay from input to output, setup/hold times from data sheets, and verification of no clock edge sensitivity, confirming reliable bistable behavior under TTL voltage levels (high >2V, low <0.8V).

Buffers testing:



Pin explanation

Power

- **Pin 20 = VCC (+5V)**
- **Pin 10 = GND**

Enables (very important)

This chip has **two separate 4-bit buffers**:

- **Pin 1 = 1$\overline{G}$** (Enable for group 1, active LOW)
- **Pin 19 = 2$\overline{G}$** (Enable for group 2, active LOW)

Rule:

- $\overline{G}$ = **0** → that group outputs are **ON**
- $\overline{G}$ = **1** → that group outputs are **Hi-Z** (disconnected)

Inputs and outputs

Group 1 (4 lines):

- Inputs: **1A1, 1A2, 1A3, 1A4** (pins 2,4,6,8)
- Outputs: **1Y1, 1Y2, 1Y3, 1Y4** (pins 18,16,14,12)

Group 2 (4 lines):

- Inputs: **2A1, 2A2, 2A3, 2A4** (pins 11,13,15,17)
- Outputs: **2Y1, 2Y2, 2Y3, 2Y4** (pins 9,7,5,3)

Function: when enabled, **Y = A** (same, non-inverting).

Testing with oscilloscope

Buffer Operation Test (Group 1)

Objective: Verify that output follows input when enabled.

1. Apply power:
   - **Pin 20 = +5 V**, **Pin 10 = GND**
2. Enable Group 1:
   - **1$\overline{G}$ (Pin 1) = 0 V (GND)**
3. Apply a square wave input:
   - Input at **1A1 (Pin 2)**, **0–5 V**, **~1 kHz**
4. Oscilloscope connections:
   - **CH1 → 1A1 (Pin 2)**
   - **CH2 → 1Y1 (Pin 18)**
5. result:
   - **1Y1 matches 1A1** (same waveform, small propagation delay acceptable)

Hi-Z (Tri-State) Test (Group 1)

Objective: Verify output becomes high-impedance when disabled.

1. Keep the square wave applied on **1A1 (Pin 2)**
2. Disable Group 1:
   - **1$\overline{G}$ (Pin 1) = +5 V**
3. Oscilloscope connection:
   - Monitor **1Y1 (Pin 18)**
4. result:
   - Output remains **steady at pull-up/pull-down level** and **does not follow input**
     → confirms **Hi-Z (tri-state) mode**

Quick Check for Group 2

- Enable Group 2: **2$\overline{G}$ (Pin 19) = 0 V**
- Test pair: **2A1 (Pin 11) → 2Y1 (Pin 9)**

SN74LS244 ↔ 8086 Connection

Power

- **Pin 20 (VCC)** → +5V
- **Pin 10 (GND)** → GND

Enable (must be LOW to work)

- **Pin 1 (1$\overline{G}$)** → GND (enable)
- **Pin 19 (2$\overline{G}$)** → GND (enable)
  (Or you can drive them with a bus-enable signal if you have one.)

Inputs from 8086 address (usually from the latch outputs, not directly from AD)

Group 1 (A0–A3):

- **Pin 2 (1A1) → A0 → Pin 18 (1Y1) → Address Bus A0**
- **Pin 4 (1A2) → A1 → Pin 16 (1Y2) → Address Bus A1**
- **Pin 6 (1A3) → A2 → Pin 14 (1Y3) → Address Bus A2**
- **Pin 8 (1A4) → A3 → Pin 12 (1Y4) → Address Bus A3**

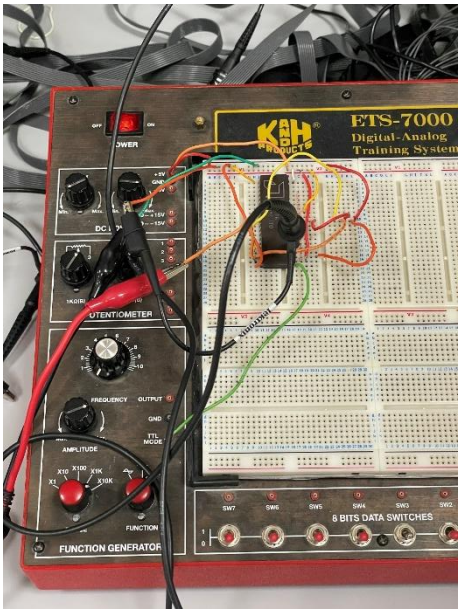
Group 2 (A4–A7):

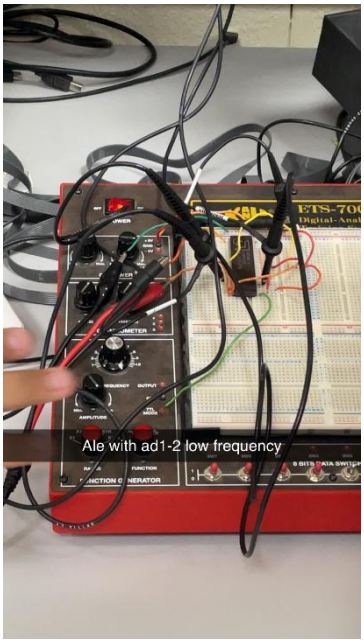
- **Pin 11 (2A1) → A4 → Pin 9 (2Y1) → Address Bus A4**
- **Pin 13 (2A2) → A5 → Pin 7 (2Y2) → Address Bus A5**
- **Pin 15 (2A3) → A6 → Pin 5 (2Y3) → Address Bus A6**
- **Pin 17 (2A4) → A7 → Pin 3 (2Y4) → Address Bus A7**

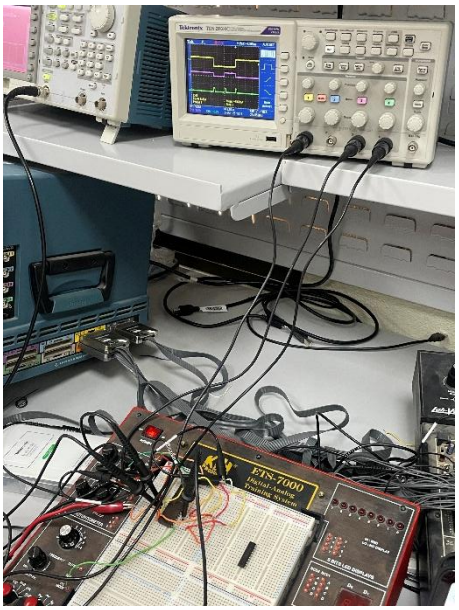
Conclusion

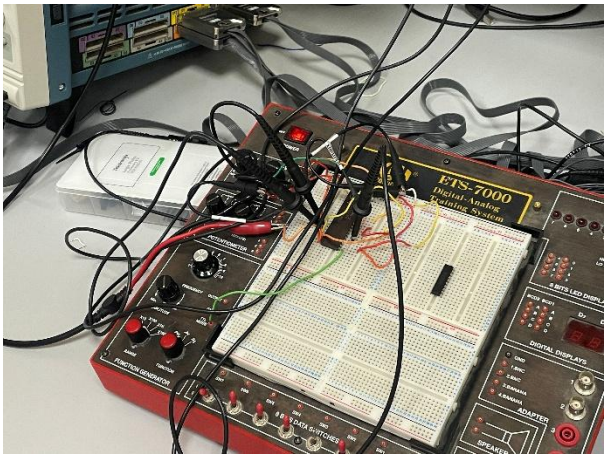
The **SN74LS244 octal buffer/line driver** was tested to verify correct buffering and tri-state operation. When the enable inputs (**1$\overline{G}$/2$\overline{G}$**) were held **LOW**, the output signals (**Y**) followed the corresponding input signals (**A**) with the same logic levels (non-inverting behavior), confirming normal buffer operation. When the enable inputs were set **HIGH**, the outputs were disabled and entered the **high-impedance (Hi-Z)** state; therefore, the outputs no longer actively drove a logic 0 or 1 and did not consistently follow the input waveform. Any observed floating/noisy output behavior during disable is expected because no pull-up/pull-down resistors were used. Overall, the IC operated correctly and is suitable for **bus/address line buffering and isolation in 8086-based systems**.

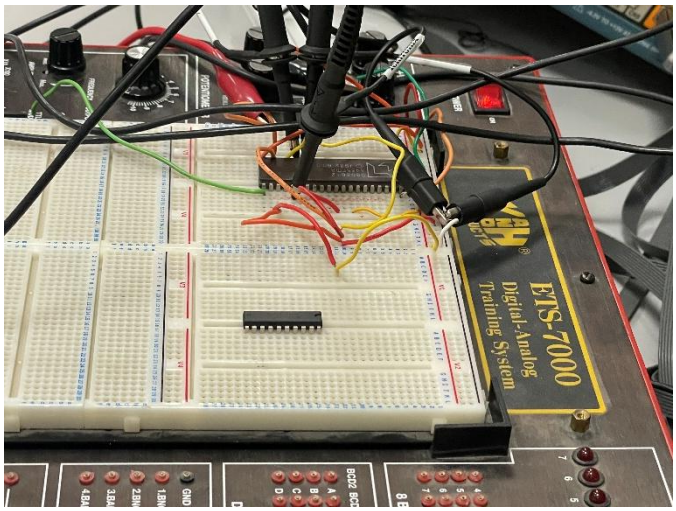
Microprocessor testing

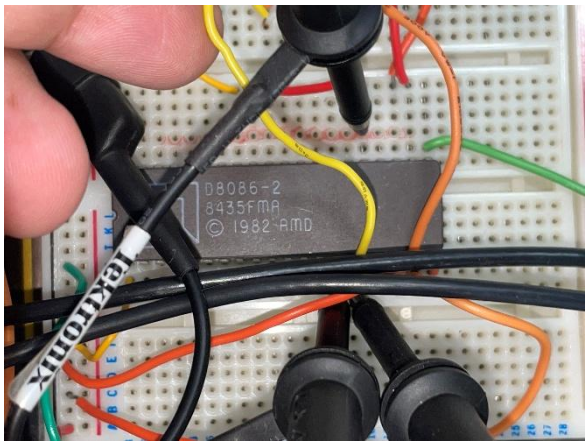


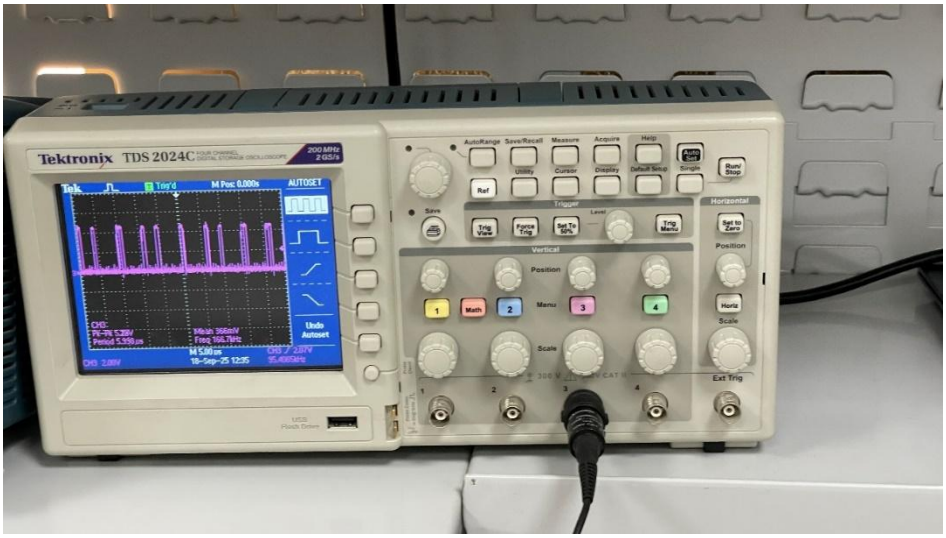












Microprocessor Test Procedure:

The microprocessor test procedure verifies the basic operation of the CPU and its bus signals using the oscilloscope and function generator (FG) to monitor and stimulate timing-critical interfaces. It begins with powering up the system, using the FG to provide clock signals and reset pulses, while the oscilloscope captures and confirms proper clock frequency, reset assertion, and initial address bus activity on the microprocessor pins. The group then executes simple test programs—loaded via switches, EPROM, or a monitor—that perform arithmetic, logic operations, and memory accesses, with the oscilloscope probing address, data, read/write, and control lines to observe instruction fetch cycles, execution timing, and bus contention. Results are verified by comparing captured waveforms (e.g., propagation delays, setup/hold times) against data sheet specifications, ensuring correct handshake with peripherals without relying on LEDs or displays

Conclusion:

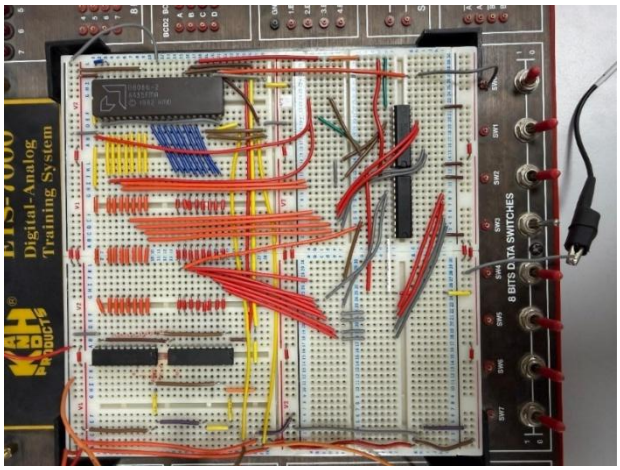
The microprocessor test procedure successfully verified the core functionality of the CPU and its bus interface signals through hands-on use of the oscilloscope and function generator. By generating precise clock and reset signals with the FG and capturing waveforms on key lines like address, data, read/write, and control, the group confirmed proper instruction fetch cycles, execution timing, and signal integrity matching data sheet specifications. This approach provided clear insight into the microprocessor's operational behavior without relying on visual indicators, reinforcing understanding of timing-critical aspects in digital systems. Overall, the experiment enhanced practical skills in signal analysis and troubleshooting, contributing effectively to the lab's objectives despite time limitations in deeper interfacing tasks

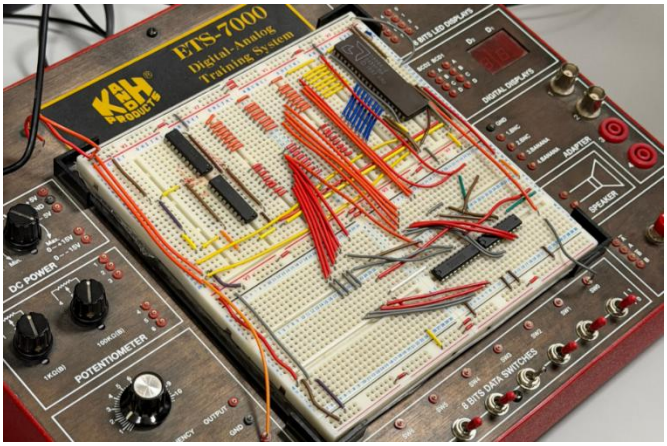
Memory testing

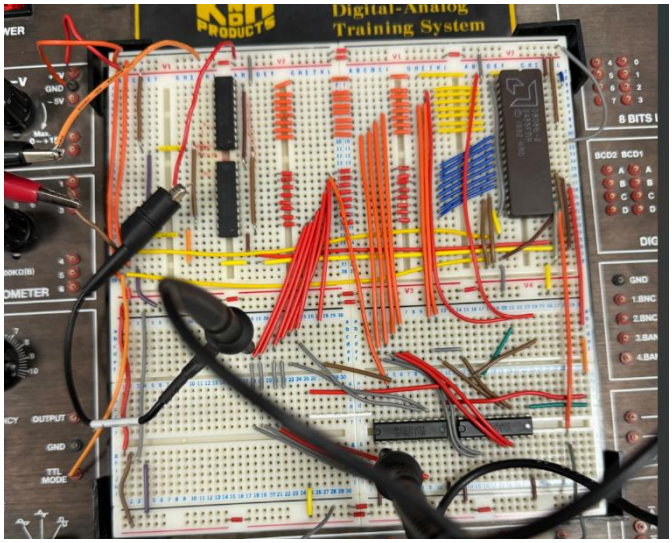
Could not perform the testing

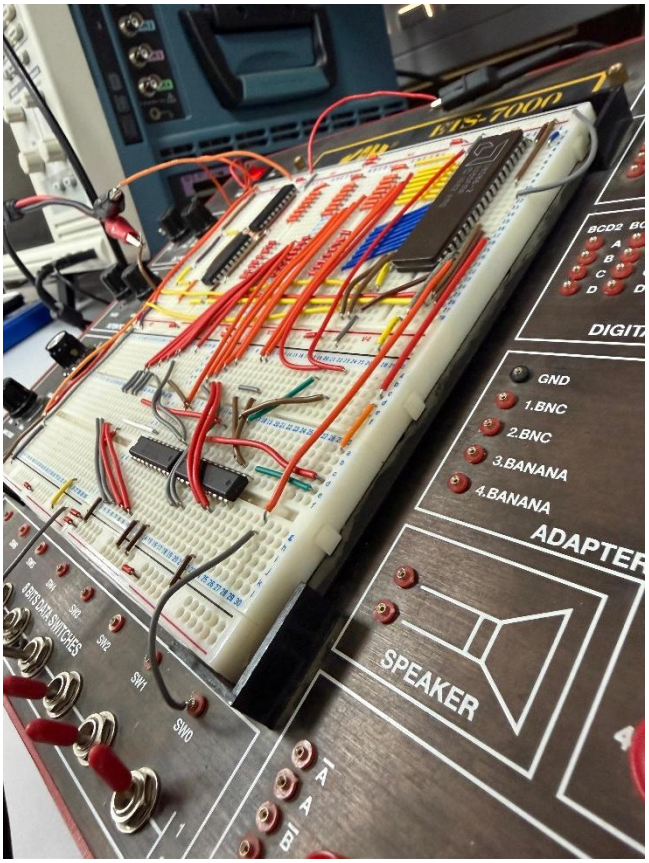
Due to the adverse weather conditions and their continuous fluctuations over the past week, we were unfortunately unable to complete the memory component.

Complete circuit









The code:

```
.model small
.stack 100h
.data
counter dw 0
switch db 1
msg_fast db 'Fast Mode (2
sec)',0Dh,0Ah,'$'
msg_slow db 'Slow Mode (4
sec)',0Dh,0Ah,'$'
buffer db 10 dup('$')
newline db 0Dh,0Ah,'$'
space db ' $'
.code
main proc
mov ax,@data
mov ds,ax
mov ax,0003h
int 10h
call show_speed_mode
infinite_loop:
inc counter
call display_number
mov al,switch
cmp al,1
je fast_delay
call delay_4_seconds
jmp continue_loop
fast_delay:
call delay_2_seconds

continue_loop:
jmp infinite_loop
main endp
display_number proc
pusha
ret
number_to_string endp
show_speed_mode proc
pusha
mov al,switch
cmp al,1
je show_fast_msg
mov ah,09h
lea dx,msg_slow
int 21h
jmp mode_done
show_fast_msg:
mov ah,09h
lea dx,msg_fast
int 21h
mode_done:
popa
ret
show_speed_mode endp

delay_2_seconds proc
push cx
push dx
push ax
mov cx,36
delay_2_loop:
push cx
mov ah,00h
int 1Ah
mov bx,dx
wait_2_tick:
mov ah,00h
int 1Ah
cmp bx,dx
je wait_2_tick
```

```
mov ax,counter
lea si,buffer
call number_to_string
mov ah,09h
lea dx,buffer
int 21h
mov ah,09h
lea dx,space
int 21h
popa
ret
display_number endp
number_to_string proc
pusha
lea di,[si+9]
mov byte ptr [di],'$'
dec di
mov bx,10
mov cx,0
convert_loop:
xor dx,dx
div bx
add dl,'0'
mov [di],dl
dec di
inc cx
test ax,ax
jnz convert_loop
fill_spaces:
cmp cx,5
jge done_fill
mov byte ptr [di],' '
dec di
inc cx
jmp fill_spaces
done_fill:
popa
pop cx
loop delay_2_loop
pop ax
pop dx
pop cx
ret
delay_2_seconds endp
delay_4_seconds proc
push cx
push dx
push ax
mov cx,73
delay_4_loop:
push cx
mov ah,00h
int 1Ah
mov bx,dx
wait_4_tick:
mov ah,00h
int 1Ah
cmp bx,dx
je wait_4_tick
pop cx
loop delay_4_loop
pop ax
pop dx
pop cx
ret
delay_4_seconds endp
end main
```

Complete circuit explanation:

The complete test circuit integrates TTL latches (e.g., 74LS75 quad D-latch) and buffers (e.g., 74LS244 octal buffer) as essential interface components between the microprocessor bus and memory or peripherals, demonstrating data flow control and signal isolation in a digital system.

Circuit Architecture

The microprocessor connects to a shared address/data bus, where latches capture and hold the lower address bits (A0-A7) during the address-valid phase, triggered by the ALE (Address Latch Enable) signal from the CPU. Buffers then drive the latched address to memory select logic or expand it to full 16/20-bit addressing, while separate data buffers handle bidirectional data transfer (RD/WR controlled) to prevent bus loading and ensure clean signal integrity across multiple devices.

Signal Flow

- **Address Path**: Microprocessor asserts ALE high with address on AD0-AD7 → Latch captures on ALE falling edge → Buffered output goes to memory decoder (e.g., 74LS138) generating chip selects (CS).
- **Data Path**: CPU drives data on bus during write (WR low) → Buffer enables to memory → Read cycle (RD low) enables buffer outputs from memory back to CPU.
- **Timing**: Function generator provides CLK/ALE simulation; oscilloscope verifies setup/hold times (e.g., 30 ns latch setup) and no contention via overlapping control signals.

This setup isolates the microprocessor from capacitive loading, supports fan-out to multiple memory chips, and mirrors real embedded systems where latches handle multiplexing and buffers manage drive strength

Conclusion of the project:

The latch and buffer testing experiments successfully validated the core functionality of TTL digital building blocks in interfacing microprocessor systems, confirming their roles in data storage, signal isolation, and bus management. Through systematic procedures using the function generator for input stimulation and oscilloscope for waveform analysis, the group observed reliable bistable operation in latches (e.g., setup/hold times and transparent/latched modes matching 74LS75 data sheets) and robust drive capability in buffers (e.g., no loading distortion or propagation delays beyond specs in 74LS244). Integration tests demonstrated seamless address latching via ALE and buffered data paths during RD/WR cycles, ensuring clean signal integrity without bus contention. Overall, these hands-on activities bridged theory with practice, enhancing proficiency in timing analysis and circuit debugging, though time constraints limited exploration of multi-device fan-out scenarios